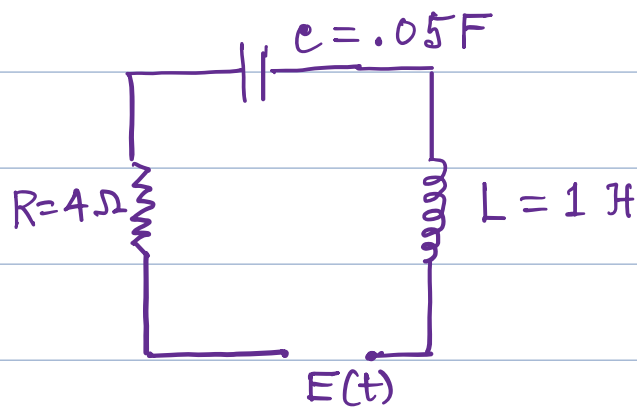


MTH 204 : Lecture 20



(we can write $E(t) = 34e^{-t} [1 - u(t-4)]$ for $t > 0$)

Find the response (the current) of the RLC circuit where $E(t)$ is given by

$$E(t) = 34e^{-t} \quad \text{for } 0 < t < 4$$

$$= 0 \quad \text{for } t > 4$$

current and charge are initially zero.

Now

$$L i'(t) + R i(t) + \frac{1}{C} \int_0^t i(\tau) d\tau = E(t) \quad \left(\begin{array}{l} \text{Also } i(0) = 0 \\ i'(0) = 0 \end{array} \right)$$

Taking Laplace transform:

$$L s I + R I + \frac{1}{C s} I = \frac{34}{s+1} - \frac{34 e^{-4s-4}}{s+1}$$

Note that $\mathcal{L}\{e^{-t} u(t-4)\} = \mathcal{L}\{e^{-4} [e^{-(t-4)} u(t-4)]\}$

$$= e^{-4} \mathcal{L}\{e^{-(t-4)} u(t-4)\} = e^{-4} e^{-4s} \mathcal{L}\{e^{-t}\}$$

$$= \frac{e^{-4s-4}}{s+1}$$

$$\Rightarrow SI + 4I + \frac{20}{s} I = \frac{34}{s+1} - \frac{34 e^{-4s-4}}{s+1}$$

$$\Rightarrow \frac{(s^2+4s+20)}{s} I = \frac{34}{s+1} - \frac{34 e^{-4s-4}}{s+1}$$

$$\Rightarrow I = \frac{34s}{(s+1)(s^2+4s+20)} (1 - e^{-4s-4})$$

$$\Rightarrow i = \mathcal{L}^{-1} \left\{ \frac{34s}{(s+1)(s^2+4s+20)} - e^{-4} \frac{34s e^{-4s}}{(s+1)(s^2+4s+20)} \right\}$$

$$\text{Now let } \frac{34s}{(s+1)(s^2+4s+20)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+4s+20}$$

$$\Rightarrow 34s = As^2 + 4As + 20A + Bs^2 + Cs + Bs + C$$

$$\Rightarrow A+B=0, \quad 4A+B+C=34, \quad 20A+C=0$$

$$\Rightarrow C = -20A \Rightarrow 4A+B-20A=34 \Rightarrow -16A+B=34$$

$$A+B=0$$

$$\begin{array}{r} -17A = 34 \\ \Rightarrow \boxed{A = -2} \end{array}$$

$$\text{Then } B = -A = -(-2) = \boxed{2} \text{ and } \boxed{C = 40}$$

$$\text{So, } \frac{34s}{(s+1)(s^2+4s+20)} = \frac{-2}{s+1} + \frac{2s+40}{(s+2)^2+16}$$

$$\text{Then } \mathcal{L}^{-1} \left\{ \frac{34s}{(s+1)(s^2+4s+20)} \right\} = -2 \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} + 2 \mathcal{L}^{-1} \left\{ \frac{(s+2)}{(s+2)^2+4^2} \right\} + \mathcal{L}^{-1} \left\{ \frac{36}{(s+2)^2+4^2} \right\}$$

$$= -2e^{-t} + 2e^{-2t} \cos(4t) + 9e^{-2t} \sin(4t)$$

$$\text{Therefore } \mathcal{L}^{-1} \left\{ e^{-4} e^{-4s} \frac{34s}{(s+1)(s^2+4s+20)} \right\}$$

$$= e^{-4} \left[-2e^{-(t-4)} + 2e^{-2(t-4)} \cos\{4(t-4)\} + 9e^{-2(t-4)} \sin\{4(t-4)\} \right] u(t-4)$$

$$= \left[-2e^{-t} + 2e^{-2(t-2)} \cos\{4(t-4)\} + 9e^{-2(t-2)} \sin\{4(t-4)\} \right] u(t-4)$$

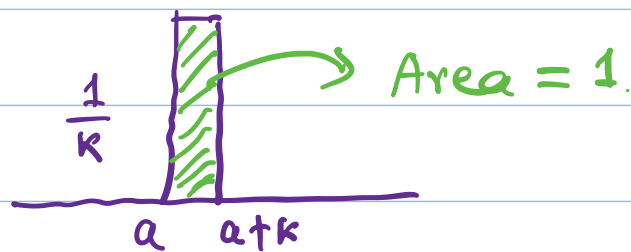
$$\text{So, } i(t) = \mathcal{L}^{-1} \left\{ \frac{34s}{(s+1)(s^2+4s+20)} \left(1 - e^{-4s-4} \right) \right\}$$

$$= \left[-2e^{-t} + e^{-2t} \{ 2\cos(4t) + 9\sin(4t) \} - u(t-4) \left[-2e^{-t} + e^{-2(t-2)} \{ 2\cos\{4(t-4)\} + 9\sin\{4(t-4)\} \} \right] \right]$$

Short Impulses. Dirac Delta Function

Dirac Delta function represent short impulses and they can be seen as the limit of the function

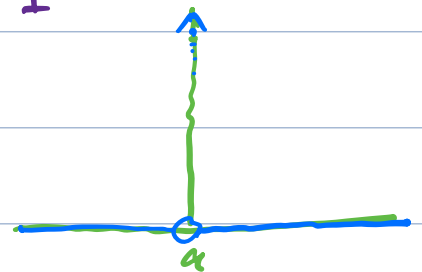
$$f_k(t-a) = \begin{cases} \frac{1}{k} & a \leq t \leq a+k \\ 0 & \text{otherwise} \end{cases}$$



Note that for any k , the area under f_k is always 1.

$$I_k = \int_0^{\infty} f_k(t-a) dt = \int_a^{a+k} \frac{1}{k} dt = \frac{1}{k} (k+a-a) = 1$$

$$\delta(t-a) = \lim_{k \rightarrow 0} f_k(t-a)$$



Note: Technically δ is not a function
It is a generalized function (distribution)

It is zero everywhere except at a

point whose value is ∞ and its integral is 1.

It has the shifting property

$$\int_{-\infty}^{\infty} g(t) \delta(t-a) dt = g(a)$$

Laplace Transform of $\delta(t-a)$:

$$f_k(t-a) = \frac{1}{k} [u(t-a) - u(t-(a+k))]$$

$$\mathcal{L}\{f_k(t-a)\} = \mathcal{L}\left\{\frac{1}{k} [u(t-a) - u(t-(a+k))]\right\}$$

$$= \frac{1}{ks} [e^{-as} - e^{-(a+k)s}]$$

$$= e^{-as} \frac{(1 - e^{-ks})}{ks}$$

Taking limit as $k \rightarrow 0$ and Using L'Hospital Rule

$$\boxed{\mathcal{L}\{\delta(t-a)\} = e^{-as}}$$

$$\left(\begin{aligned} &\lim_{k \rightarrow 0} \frac{1 - e^{-ks}}{ks} \\ &= \lim_{k \rightarrow 0} \frac{\cancel{1} e^{-ks}}{\cancel{s}} = \lim_{k \rightarrow 0} e^{-ks} \\ &= 1 \end{aligned} \right)$$

Ex: Hammer blow on a mass spring system:

What is the impulse response of a mass-spring system to a blow at $t=1$?

$$y'' + 4y' + 5y = \delta(t-1)$$

$$(y(0) = 0, y'(0) = 3)$$

Taking the Laplace transform:

$$\{s^2 Y - sy(0) - y'(0)\} + 4\{sY - y(0)\} + 5Y = e^{-s}$$

$$\Rightarrow s^2 Y - 3 + 4sY + 5Y = e^{-s}$$

$$\Rightarrow s^2 Y + 4sY + 5Y = 3 + e^{-s}$$

$$\Rightarrow Y = \frac{3 + e^{-s}}{s^2 + 4s + 5}$$

$$\Rightarrow Y = \frac{3}{s^2 + 4s + 5} + \frac{e^{-s}}{s^2 + 4s + 5}$$

$$\Rightarrow Y = \frac{3}{(s+2)^2 + 1} + \frac{e^{-s}}{(s+2)^2 + 1}$$

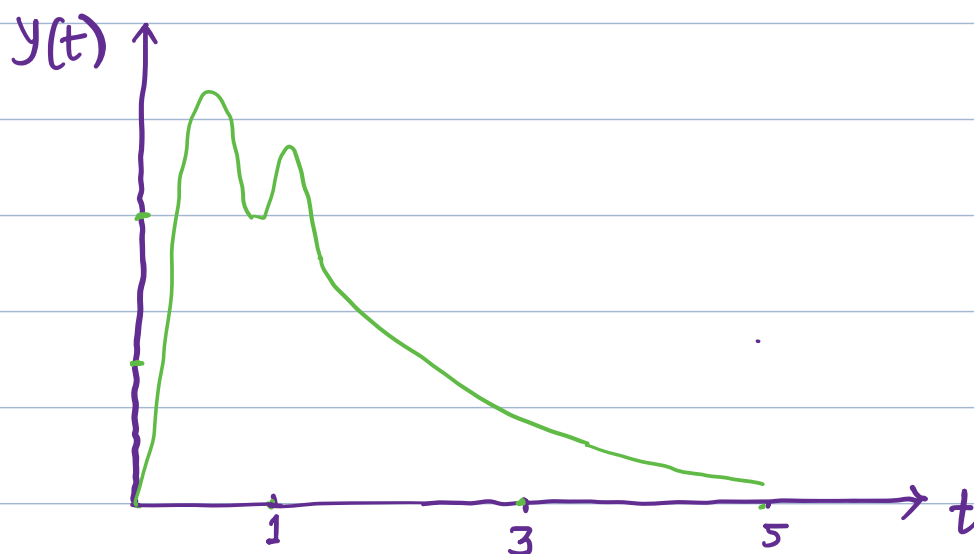
$$\text{So, } y = \mathcal{L}^{-1} \left\{ \frac{3}{(s+2)^2 + 1} \right\} + \mathcal{L}^{-1} \left\{ \frac{e^{-s}}{(s+2)^2 + 1} \right\}$$

$$= 3e^{-2t} \sin t + u(t-1) e^{-2(t-1)} \sin(t-1)$$

$$\text{So, } y(t) = 3e^{-2t} \sin t \quad \text{if } 0 \leq t < 1$$

$$= 3e^{-2t} \sin t + e^{-2(t-1)} \sin(t-1) \quad \text{if } t \geq 1$$

(impulse)



Mass-spring System Under a Square Wave

The response of the damped mass-spring system under a square wave is modeled

by $y'' + 4y' + 5y = r(t) = u(t-1) - u(t-2)$
 $y(0) = 0, y'(0) = 0$ (square wave)

Taking Laplace Transform of both sides,

$$s^2 Y + 4sY + 5Y = \frac{1}{s} e^{-s} - \frac{1}{s} e^{-2s} \Rightarrow Y = \frac{e^{-s} - e^{-2s}}{s(s^2 + 4s + 5)}$$

Let $\frac{1}{s(s^2 + 4s + 5)} = \frac{A}{s} + \frac{Bs + c}{s^2 + 4s + 5} = \frac{As^2 + 4As + 5A + Bs^2 + cs}{s(s^2 + 4s + 5)}$

$$\Rightarrow A + B = 0, 4A + c = 0, 5A = 1 \Rightarrow A = \frac{1}{5}, B = -\frac{1}{5}, c = -\frac{4}{5}$$

So, $\frac{1}{s(s^2 + 4s + 5)} = \frac{1}{5s} + \frac{-\frac{1}{5}s - \frac{4}{5}}{(s+2)^2 + 1} = \frac{1}{5s} + \frac{-\frac{1}{5}s - \frac{2}{5} - \frac{2}{5}}{(s+2)^2 + 1}$

Therefore $\mathcal{L}^{-1} \left\{ \frac{1}{s(s^2 + 4s + 5)} \right\} = \frac{1}{5} - \frac{1}{5} e^{-2t} \cos t - \frac{2}{5} e^{-2t} \sin t$

Hence $y(t) = \mathcal{L}^{-1} \left\{ \frac{e^{-s}}{s(s^2 + 4s + 5)} - \frac{e^{-2s}}{s(s^2 + 4s + 5)} \right\}$

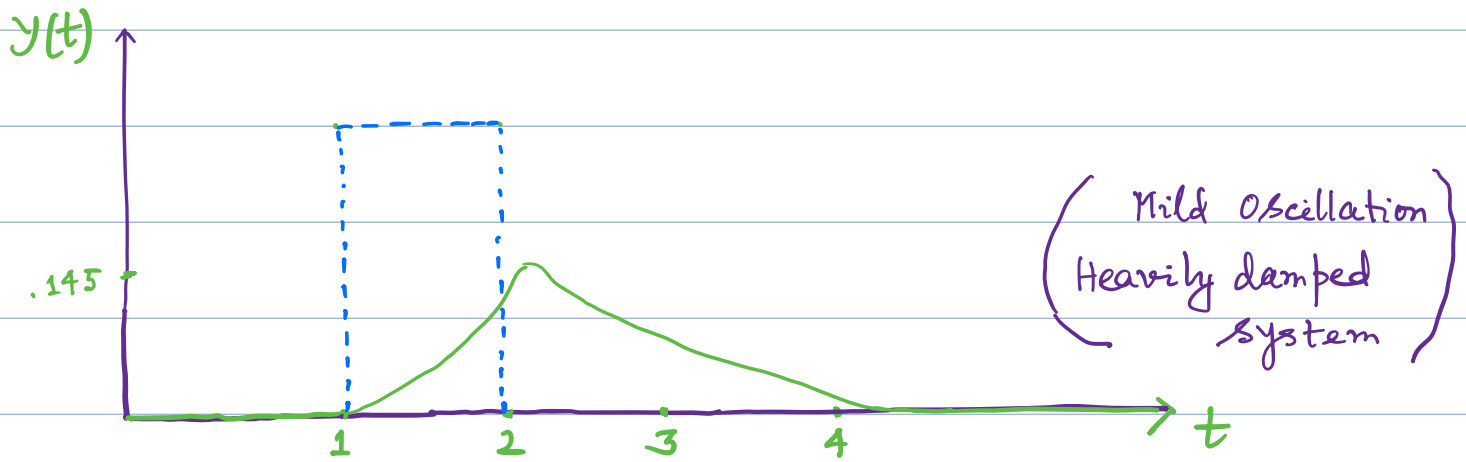
$$= \left[\frac{1}{5} - \frac{1}{5} e^{-2(t-1)} \cos(t-1) - \frac{2}{5} e^{-2(t-1)} \sin(t-1) \right] u(t-1)$$

$$- \left[\frac{1}{5} - \frac{1}{5} e^{-2(t-2)} \cos(t-2) - \frac{2}{5} e^{-2(t-2)} \sin(t-2) \right] u(t-2)$$

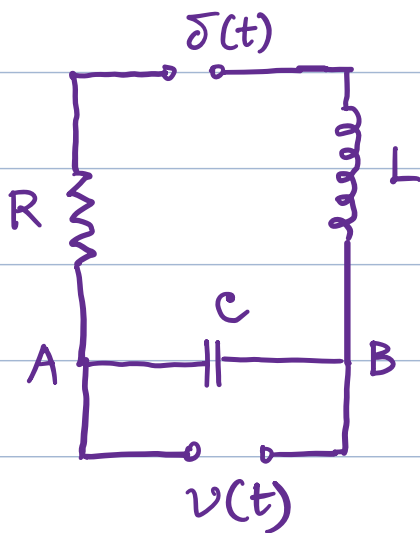
So, $y(t) = 0$ if $0 \leq t < 1$

$$= \frac{1}{5} - \frac{1}{5} e^{-2(t-1)} \cos(t-1) - \frac{2}{5} e^{-2(t-1)} \sin(t-1) \text{ if } 1 \leq t < 2$$

$$= \frac{1}{5} \left\{ e^{-2(t-2)} \cos(t-2) - e^{-2(t-1)} \cos(t-1) \right\} \\ + \frac{2}{5} \left\{ e^{-2(t-2)} \sin(t-2) - e^{-2(t-1)} \sin(t-1) \right\} \\ \text{if } t \geq 2$$



Ex: Four terminal RLC-Network



Find the output voltage response if $R = 20 \Omega$, $L = 1 \text{ H}$, $C = 10^{-4} \text{ F}$.

The input is $\delta(t)$ (a unit impulse at time $t=0$)

The current and charge are zero at time $t=0$

If $i(t)$ is the current at time t , then

$$L i' + R i + \frac{q}{C} = \delta(t) \\ \left(\infty \quad L i' + R i + \frac{1}{C} \int_0^t i(t) dt = \delta(t) \right)$$

Note that $i(t) = q' = \frac{dq}{dt} = c \frac{dv}{dt}$ $\left(\frac{q}{c} = v\right)$

Hence $LCv'' + RCv' + v = \delta(t)$

$$\Rightarrow v'' + \frac{R}{L}v' + \frac{1}{LC}v = \frac{1}{LC}\delta(t)$$

Note that $v(0) = 0$, $v'(0) = 0$

Taking Laplace Transform of both sides

$$s^2 \mathcal{L}(v) + \frac{R}{L} s \mathcal{L}(v) + \frac{1}{LC} \mathcal{L}(v) = \frac{1}{LC}$$

$$\left(\text{Since } \mathcal{L}(\delta(t)) = e^{0 \times t} = 1\right)$$

Now $R = 20 \Omega$, $C = 10^{-4} F$, $L = 1 H$

$$\Rightarrow s^2 \mathcal{L}(v) + 20s \mathcal{L}(v) + 10^4 \mathcal{L}(v) = 10^4$$

$$\Rightarrow \mathcal{L}(v) = \frac{10^4}{s^2 + 20s + 10^4}$$

$$\Rightarrow \mathcal{L}(v) = \frac{10^4}{(s+10)^2 + 9900}$$

$$\Rightarrow \mathcal{L}(v) = \frac{10^4 (\sqrt{9900})}{\sqrt{9900} \{ (s+10)^2 + (\sqrt{9900})^2 \}}$$

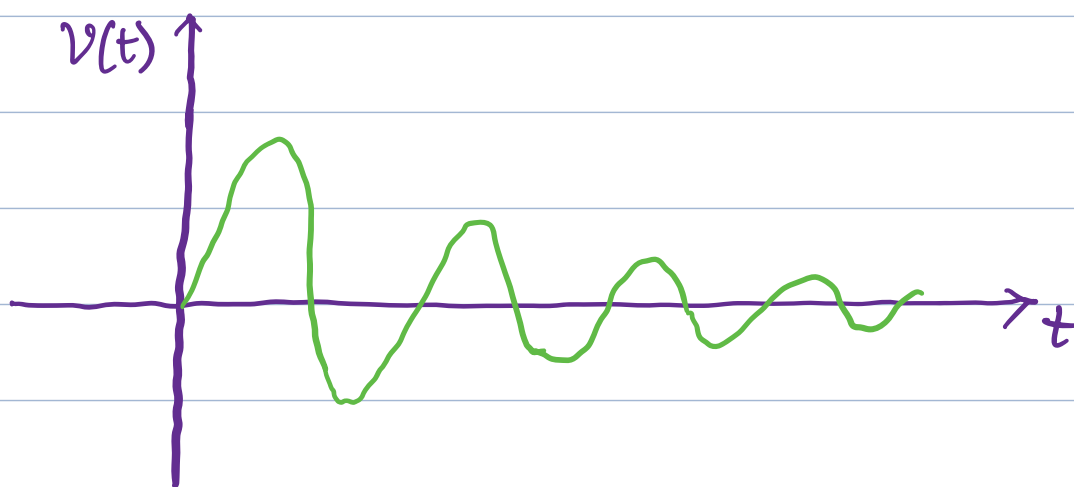
Taking the inverse Laplace Transform, we

get

$$v(t) = \frac{10^4}{\sqrt{9900}} \sin(\sqrt{9900} t) \times e^{-10t}$$

or

$$v(t) = 100.5 e^{-10t} \sin(99.5t)$$



Partial Fractions :

In many cases, the solution Y of a subsidiary equation usually appears as a quotient of polynomials

$$Y(s) = \frac{F(s)}{G(s)}$$

Depending on the factorization of $G(s)$,

we have a partial fraction representation leading to a sum of expressions whose inverse we can obtain.

These representations are sometimes called Heaviside Expansions.

- An unpeated factor $(s-a)$ in $G(s)$
 - Single partial fraction $\frac{A_1}{s-a}$
 - Inverse: $A_1 e^{-at}$
- Repeated real factor $(s-a)^2 \rightarrow \frac{A_2}{(s-a)^2} + \frac{A_1}{(s-a)}$
 - Inverse : $(A_1 + A_2 t) e^{-at}$
- Repeated real factor $(s-a)^3 \rightarrow \frac{A_3}{(s-a)^3} + \frac{A_2}{(s-a)^2} + \frac{A_1}{(s-a)}$
 - Inverse : $(A_1 + A_2 t + \frac{1}{2} A_3 t^2) e^{-at}$
- Unpeated complex factors $(s-a)(s-\bar{a})$
where $a = \alpha + i\beta$
 - $\frac{A_1 s + A_2}{(s-\alpha)^2 + \beta^2} = \frac{A_1(s-\alpha) + A_2 + A_1 \alpha}{(s-\alpha)^2 + \beta^2}$

$$\rightarrow \text{Inverse: } e^{-\alpha t} \left\{ A_1 \cos(\beta t) + \frac{A_2 + A_1 \alpha}{\beta} \sin(\beta t) \right\}$$

Ex: $y'' + 2y' + 2y = 10 \sin(2t) (u(t) - u(t - \pi))$
 $y(0) = 1, y'(0) = -5$

Taking Laplace Transform:

$$(s^2 Y - s + 5) + 2(sY - 1) + 2Y$$

$$= 10 \frac{2}{s^2 + 4} (1 - e^{-\pi s})$$

$$\Rightarrow (s^2 + 2s + 2)Y = (s - 3) + \frac{20}{s^2 + 4} (1 - e^{-\pi s})$$

$$\Rightarrow Y = \frac{(s - 3)}{(s^2 + 2s + 2)} + \frac{20}{(s^2 + 2s + 2)(s^2 + 4)} (1 - e^{-\pi s})$$

$$\text{Now } \mathcal{L}^{-1} \left(\frac{(s - 3)}{s^2 + 2s + 2} \right) = \mathcal{L}^{-1} \left\{ \frac{(s + 1) - 4}{(s + 1)^2 + 1} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{s + 1}{(s + 1)^2 + 1} \right\} - \mathcal{L}^{-1} \left\{ \frac{4}{(s + 1)^2 + 1} \right\}$$

$$= e^{-t} \cos(t) - 4e^{-t} \sin t$$

$$= e^{-t} [\cos(t) - 4 \sin(t)]$$

Now let $\frac{20}{(s^2+2s+2)(s^2+4)} = \frac{As+B}{(s+1)^2+1} + \frac{Ms+N}{s^2+4}$

Then $20 = (As+B)(s^2+4) + (Ms+N)(s^2+2s+2)$

$$\Rightarrow 20 = (A+M)s^3 + (B+N+2M)s^2 + (4A+2M+2N)s + (4B+2N)$$

$$\Rightarrow A+M=0, B+N+2M=0, 4A+2M+2N=0$$

and $4B+2N=20$

$$\Rightarrow M=-A, N=10-2B \Rightarrow B+10-2B-2A=0$$

$$\Rightarrow 2A+B=10 \quad \text{and} \quad 4A-2A+20-4B=0$$

$$\Rightarrow \left. \begin{array}{l} 2A-4B=-20 \\ \text{and } 2A+B=10 \end{array} \right\}$$

Hence $-5B=-30 \Rightarrow \boxed{B=6}$

Now $2A=10-6 \Rightarrow \boxed{A=2}$

Then $\boxed{M=-2}$ and $\boxed{N=-2}$

Therefore

$$\frac{20}{(s^2+2s+2)(s^2+4)} = \frac{2s+6}{(s+1)^2+1} - 2 \frac{(s+1)}{s^2+4}$$

$$= \frac{2(s+1)+4}{(s+1)^2+1} - 2 \frac{s}{s^2+4} - 2 \frac{1}{s^2+4}$$

$$= 2 \frac{(s+1)}{(s+1)^2+1} + 4 \frac{1}{(s+1)^2+1} - 2 \frac{s}{s^2+2^2} - \frac{2}{s^2+2^2}$$

$$\text{So, } \mathcal{L}^{-1} \left\{ \frac{20}{(s^2+2s+2)(s^2+4)} \right\}$$

$$= e^{-t} [2 \cos t + 4 \sin t] - 2 \cos(2t) - \sin(2t)$$

Combining, we can say that for $0 < t < \pi$

$$\boxed{y(t) = 3e^{-t} \cos t - 2 \cos(2t) - \sin(2t)}$$

and for $t > \pi$

$$y(t) = 3e^{-t} \cos t - 2 \cos(2t) - \sin(2t)$$

$$- e^{-(t-\pi)} [2 \cos(t-\pi) + 4 \sin(t-\pi)]$$

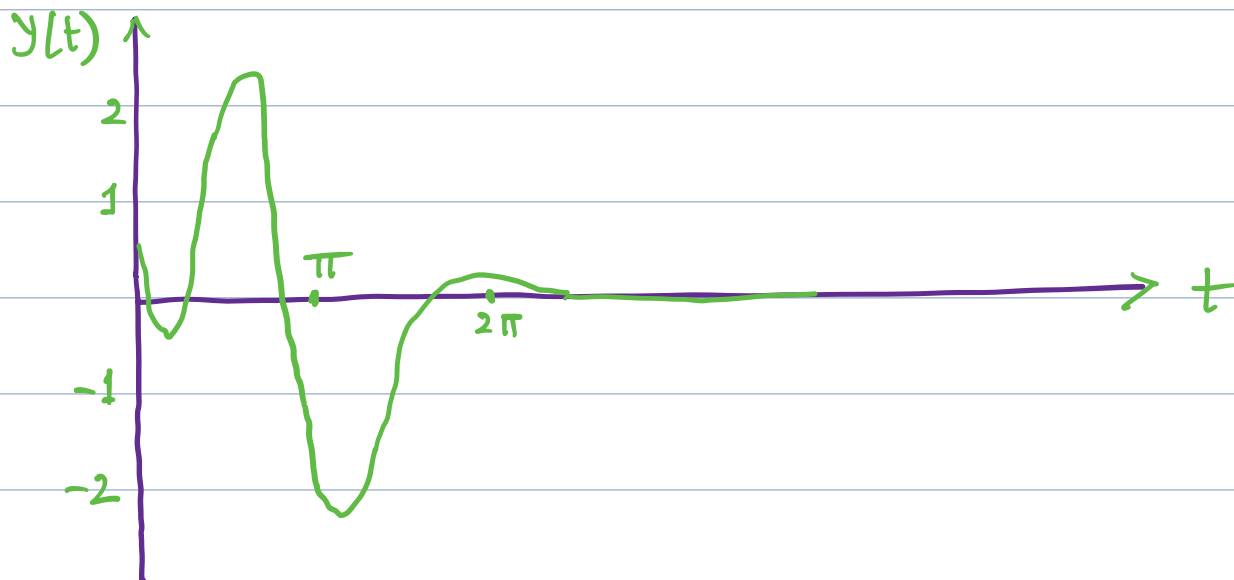
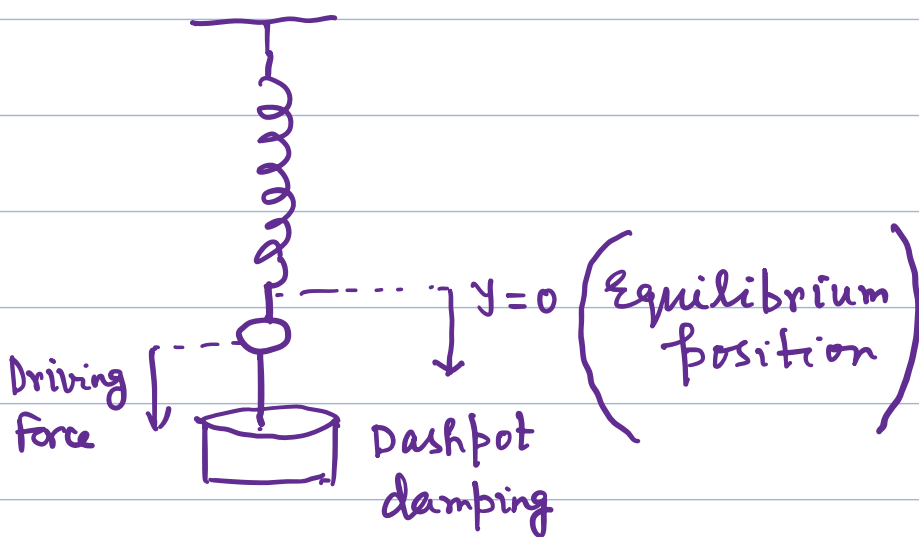
$$+ 2 \cos[2(t-\pi)] + \sin[2(t-\pi)]$$

$$\begin{aligned}
 &= 3e^{-t} \cos t - 2 \cos(2t) - \sin(2t) \\
 &\quad + e^{-(t-\pi)} [2 \cos t + 4 \sin t] \\
 &\quad + 2 \cos(2t) + \sin(2t)
 \end{aligned}$$

$$\Rightarrow y(t) = e^{-t} [(3 + 2e^{\pi}) \cos t + 4e^{\pi} \sin t] \quad \text{for } t > \pi$$

Note:

In this case differential equation for damped forced vibration is considered.



Convolution:

If two functions f and g satisfy the assumptions of the existence theorem so that their transforms F and G exist, then

$$\mathcal{L}\{f * g\}(s) = F(s)G(s)$$

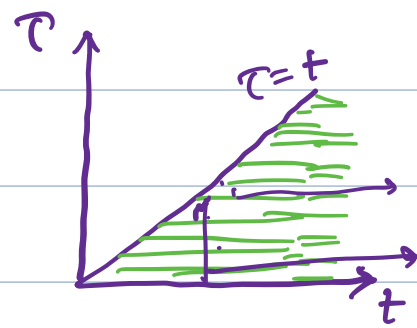
where the convolution of f and g (denoted by $f * g$) is defined by

$$(f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau$$

Proof:

$$\begin{aligned} F(s)G(s) &= \left(\int_0^\infty e^{-s\tau} f(\tau) d\tau \right) \left(\int_0^\infty e^{-sp} g(p) dp \right) \\ &= \left\{ \int_0^\infty e^{-s\tau} f(\tau) d\tau \right\} \left\{ \int_\tau^\infty e^{-s(t-\tau)} g(t-\tau) dt \right\} \quad \begin{array}{l} \text{let } t = p + \tau \\ p = t - \tau \end{array} \\ &= \left\{ \int_0^\infty e^{-s\tau} f(\tau) d\tau \right\} \left\{ e^{s\tau} \int_\tau^\infty e^{-st} g(t-\tau) dt \right\} \\ &= \int_0^\infty e^{-s\tau} e^{s\tau} f(\tau) \left(\int_\tau^\infty e^{-st} g(t-\tau) dt \right) d\tau \end{aligned}$$

$$= \int_0^{\infty} f(\tau) \left(\int_{\tau}^{\infty} e^{-st} g(t-\tau) dt \right) d\tau$$



$$= \int_0^{\infty} e^{-st} \left(\int_0^t f(\tau) g(t-\tau) d\tau \right) dt \quad \left(\text{changing the order of integration} \right)$$

$$= \mathcal{L} \left\{ \int_0^t f(\tau) g(t-\tau) d\tau \right\}$$

$$= \mathcal{L} \{ (f * g)(t) \}$$

Ex: Find $\mathcal{L}^{-1} \left\{ \frac{1}{s(s-a)} \right\}$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s(s-a)} \right\} = e^{at} * 1$$

$$\left(\begin{array}{l} \text{Since } \mathcal{L}(e^{at}) = \frac{1}{s-a} \\ \mathcal{L}(1) = \frac{1}{s} \end{array} \right)$$

$$= \int_0^t e^{a\tau} 1 d\tau = \left. \frac{e^{a\tau}}{a} \right|_0^t$$

$$= \frac{1}{a} (e^{at} - 1)$$

Ex: Find $\mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + \omega^2)^2} \right\}$

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + \omega^2)} \frac{1}{(s^2 + \omega^2)} \right\} = \left(\frac{1}{\omega} \sin \omega t \right) * \left(\frac{1}{\omega} \sin \omega t \right)$$

$$\begin{aligned}
&= \frac{1}{\omega^2} \int_0^t \sin(\omega \tau) \sin\{\omega(t-\tau)\} d\tau \\
&= \frac{1}{2\omega^2} \int_0^t [-\cos \omega t + \cos(2\omega\tau - \omega t)] d\tau \\
&= \frac{1}{2\omega^2} \left[-t \cos \omega t + \frac{\sin \omega(2\tau - t)}{2\omega} \right]_0^t \\
&= \frac{1}{2\omega^2} \left[-t \cos \omega t + \frac{\sin \omega t}{2\omega} - \frac{(-\sin \omega t)}{2\omega} \right] \\
&= \frac{1}{2\omega^2} \left[-t \cos(\omega t) + \frac{\sin(\omega t)}{\omega} \right]
\end{aligned}$$

Ex: $y'' + \omega_0^2 y = K \sin(\omega_0 t)$, $y(0) = y'(0) = 0$

Taking Laplace Transform $\left(\omega_0^2 = \frac{K}{m}, K = \text{Spring Constant} \right)$

$$s^2 Y + \omega_0^2 Y = \frac{K \omega_0}{s^2 + \omega_0^2}$$

$$\Rightarrow Y = \frac{K \omega_0}{(s^2 + \omega_0^2)^2}$$

This is a Case of Resonance. The input signal is of the same frequency as the natural

frequency of the system.

By the previous example,

$$y(t) = K\omega_0 \frac{1}{2\omega_0^2} \left[-t \cos(\omega_0 t) + \frac{\sin(\omega_0 t)}{\omega_0} \right]$$
$$= \frac{K}{2\omega_0} \left[-t \cos(\omega_0 t) + \frac{\sin(\omega_0 t)}{\omega_0} \right]$$

Properties of Convolution:

- (1) Commutativity: $f * g = g * f$
- (2) Associativity: $(f * g) * h = f * (g * h)$
- (3) Distributivity: $f * (g_1 + g_2) = f * g_1 + f * g_2$
- (4) Zero element: $f * 0 = 0 * f = 0$

Note that $f * 1 \neq f$ in general

and $(f * f)(t)$ may not be a nonnegative function.